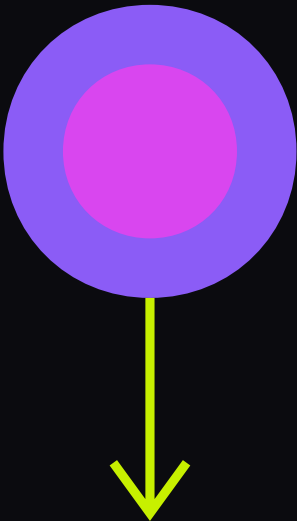


REGISTRY HIVE STAGING

Hunts source-observed reg.exe and certutil.exe staging commands for SYSTEM/SECURITY hives using the campaign's pc-* filenames.

HUNTING QUERY

CONFIDENCE: MEDIUM



DeviceProcessEvents

Campaign-specific artifact

EVIDENCE BEFORE ALERTING

REPORT DATE

12 September 2026

STATUS

Untested implementation sketch

PRIMARY PLATFORM

Microsoft Defender XDR

OPERATIONAL VALUE

9.4 / 10



EXECUTIVE ASSESSMENT

Decision-grade summary and evidence boundary.

ASSESSMENT

OPERATIONAL VALUE

9.4

Hunts source-observed reg.exe and certutil.exe staging commands for SYSTEM/SECURITY hives using the campaign's pc-* filenames.

CLASSIFICATION

Hunting query

Hunting query - Untested implementation sketch; Confidence Medium.

1primary table

0thresholds

0joins

0secondary rules

Source fact

GreyNoise published full reg save and certutil -encode commands.

Analytical choice

Observed paths include C:\Windows\Temp and C:\ProgramData.

Constraint

The actor changed behavior dynamically; the filenames are not durable.

Evidence boundary

A result does not prove PaperCut exploitation, hive content, credential recovery, exfiltration, DCSync, or compromise.



THREAT MECHANICS

Observed behavior and bounded analytical interpretation.

BEHAVIOR

01 Initial access

PaperCut NG/MF exploitation begins the reported intrusion.

→

02 Observed behavior

GreyNoise published full reg save and certutil -encode commands.

→

03 Detection pivot

Hunts source-observed reg.exe and certutil.exe staging commands for SYSTEM/SECURITY hives using the campaign's pc-* filenames.

Evidence layer	Supported statement
Publication	GreyNoise, 09 Sep 2026.
Observed activity	31 Aug 2026 onward; final observation date not stated.
Logic type	Campaign-specific artifact
ATT&CK	T1003.004 LSA Secrets

Hypothesis

If an attacker repeats the observed staging, process telemetry should show reg.exe saving SYSTEM/SECURITY to a documented pc-* hive or certutil.exe encoding the same artifact to .b64. Filename pair, sensitive hive and parent context reduce benign overlap.



TELEMETRY AND DATA CONTRACT

Minimum collection and field requirements.

TELEMETRY

REQUIRED

Microsoft Defender for Endpoint process creation telemetry with complete ProcessCommandLine.

OPTIONAL ENRICHMENT

DeviceFileEvents, process tree, host role, PaperCut inventory, network events and approved forensic activity.

Fields	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Time and device
FileName, FolderPath	Documented	Process image
ProcessCommandLine	Documented	Observed action and artifacts
ProcessUniqueId, ReportId	Documented	Event pivots
InitiatingProcess*	Documented	Parent context

Collection boundary

No payload, hive content, inferred success, rarity score, or undocumented field is used.



PRIMARY DETECTION LOGIC

Direct source-supported conditions and investigative output.

KQL

STATUS

Hunting query - Untested implementation sketch

KQL - UNTESTED

```
DeviceProcessEvents
| where
  (FileName =~ "reg.exe"
  and ProcessCommandLine has "save"
  and (ProcessCommandLine contains @"HKLM\SYSTEM"
    or ProcessCommandLine contains @"HKLM\SECURITY")
  and ProcessCommandLine has_any
    ("pc-sys.hiv", "pc-sec.hiv", "pc-security.hiv",
    "pc-system.hiv", "pc-sys-reg.hiv"))
  or
  (FileName =~ "certutil.exe"
  and ProcessCommandLine contains "-encode"
  and ProcessCommandLine has_any
    ("pc-sys.hiv", "pc-sec.hiv", "pc-security.hiv",
    "pc-system.hiv", "pc-sys-reg.hiv")
  and ProcessCommandLine contains ".b64")
| project Timestamp, DeviceId, DeviceName, AccountDomain,
  AccountName, FileName, FolderPath, ProcessCommandLine,
  ProcessId, ProcessUniqueId, InitiatingProcessFileName,
  InitiatingProcessCommandLine, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	AGGREGATION	STATUS
None	None	None	None	Untested



TRIAGE AND FALSE POSITIVES

Evidence needed before escalation.

WORKFLOW

1

Preserve raw event

Retain all projected identifiers and command context.

2

Confirm provenance

Identify operator, host role, ticket and management path.

3

Collect enrichment

DeviceFileEvents, process tree, host role, PaperCut inventory, network events and approved forensic activity.

4

Assess scope

Review adjacent identity, endpoint and PaperCut evidence.

Expected benign overlap

Forensic acquisition, backup/restore, incident response and lab exercises using the same filenames. Validate ticket, operator, parent, hash and host role.

Response boundary

A result does not prove PaperCut exploitation, hive content, credential recovery, exfiltration, DCSync, or compromise.



VALIDATION GATES

Controlled testing and cross-SIEM constraints.

QUALITY CONTROL

Validation steps

- 1. Run an innocuous lab reproduction and confirm full command-line capture.
- 2. Test the reg.exe and certutil.exe branches separately.
- 3. Search available retention and classify every real hit.
- 4. Confirm ProcessUniqueld and initiating-process context are populated.

Decision criteria

Success: Controlled commands match and any real results are attributable and investigable.

Rejection: Tokenization or truncation prevents matches, or the exact artifacts are common legitimate practice.

Failure: MDE is absent, DeviceProcessEvents is unavailable, or command lines are not retained.

Platform	Decision	Missing evidence
Microsoft Defender XDR	Hunting query	Target-environment execution and baseline
Splunk	No reliable implementation	Index, sourcetype, field extraction, sample event
Chronicle	No reliable implementation	Parser, UDM event type and exact field mapping



CONCLUSION AND REFERENCES

Coverage statement and document control.

FINAL

CONCLUSION

Hunting query

Hunts source-observed reg.exe and certutil.exe staging commands for SYSTEM/SECURITY hives using the campaign's pc-* filenames.

A result does not prove PaperCut exploitation, hive content, credential recovery, exfiltration, DCSync, or compromise.

MITRE ATT&CK

T1003.004 LSA Secrets

References

1. GreyNoise, 09 Sep 2026 - AI-Orchestrated Global Campaign Against PaperCut NG/MF
2. PaperCut, updated 10 Sep 2026 - NG/MF urgent security advisory
3. Microsoft Learn, updated 07 Aug 2026 - DeviceProcessEvents
4. MITRE ATT&CK - T1003.004

Document control

English dossier. No customer data. URLs are defanged in references.txt. HTML and rendering assets remain private.